

INTERNSHIP REPORT ON AUTOMOBILE OPERATIONS AND TECHNOLOGY

AS A JUNIOR INTERN AT MAHINDRA AUTOMOBILE

Submitted To

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Roll Number of Student: 25SCS1003005392
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BY

VEDANT KATARIA

CANDIDATE'S DECLARATION

I, Vedant Kataria do hereby declare that the internship report titled "Automobile Operations and Technology as a Junior Intern at Mahindra Automobile" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Vedant Kataria

Roll No.: 25SCS1003005392

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

Student Name: Vedant Kataria

Roll No.: 25SCS1003005392

INTERNSHIP COMPLETION CERTIFICATE

This page is reserved for the Internship Completion Certificate issued by Mahindra Automobile. The original certificate should be attached here before final submission.

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4. PROJECT DESCRIPTION

4.1 INTRODUCTION

An internship helps a student understand how the knowledge learned in college is used in a real working environment. The internship described in this report is structured around the role of a Junior Intern at Mahindra Automobile, with emphasis on automobile operations and basic technology practices, automobile operations, issue analysis, documentation, testing and teamwork.

During the internship, the focus is on understanding a typical automobile and technology workflow: receiving a requirement or issue, analysing it, identifying the required solution, implementing or configuring the solution, testing the result, documenting the work and communicating progress. This workflow connects academic concepts such as programming, databases, operating systems, networking and software engineering with professional practices.

This report explains my internship experience in a simple and practical way. It explains the organisation at a general level, identifies the problem addressed by the project, describes the objectives and scope, discusses tools and technologies, and presents a structured methodology and system architecture.

4.2 ORGANIZATION PROFILE

Mahindra Automobile is a automobile and mobility organisation operating as part of the Mahindra Group. It provides services across areas such as information technology, digital transformation, enterprise solutions, engineering services and business process services for organisations in different industries.

A large automobile industry environment requires collaboration between software developers, testers, analysts, infrastructure teams, project managers and business stakeholders. Work is generally organised through defined processes for requirement understanding, development, testing, deployment, support and documentation.

For a junior intern, exposure to such an environment is valuable because it develops both technical and professional skills. In addition to programming knowledge, an intern learns

the importance of version control, issue tracking, testing discipline, communication, documentation, security awareness and following organisational procedures.

This report intentionally keeps organisation-specific details general. Confidential client information, internal credentials, proprietary source code and restricted technical information should not be included in an academic report.

4.3 PROBLEM STATEMENT

In a professional software and IT environment, technical tasks must be completed systematically so that requirements are understood correctly, issues are analysed efficiently and solutions can be tested and documented. A junior team member may encounter multiple requests involving application functionality, data handling, troubleshooting, testing and documentation.

The project problem addressed in this internship is therefore the development of a structured approach for handling basic basic automobile and technology-related tasks. The work involved understanding the task, planning the steps, completing the assigned work, checking the result and maintaining simple documentation.

A key challenge is to convert a reported requirement or issue into a clear technical task, perform the appropriate analysis, develop or apply a solution, verify the result and record the outcome in a form that can be understood by other team members.

4.4 PROJECT OBJECTIVES

- To understand the working environment and responsibilities of a junior technology intern.
- To apply fundamental programming and computer-science concepts to practical tasks.
- To understand the basic automobile operations and technology life cycle and task-management workflow.
- To practise requirement analysis, problem decomposition and solution planning.
- To develop awareness of testing, debugging and validation techniques.

- To understand the role of documentation and professional communication in IT projects.
- To become familiar with commonly used development and collaboration tools.
- To improve teamwork, time management, technical communication and professional discipline.

4.5 SCOPE OF THE PROJECT

The scope of this internship is mainly educational. It covers the basic activities that a junior intern can perform with guidance from senior staff. It includes understanding requirements, analysing simple technical issues, preparing or modifying code/configuration where applicable, testing solutions, documenting observations and communicating progress.

The project also covers exposure to automobile operations and basic technology practices such as version control, debugging, test execution and issue tracking. The emphasis is on learning a repeatable professional process rather than claiming independent responsibility for production systems.

The scope excludes confidential client data, production credentials, proprietary architecture details, restricted source code, security-sensitive information and any activity requiring unauthorised access. Such information should remain within the organisation's approved systems and policies.

4.6 TECHNOLOGIES AND TOOLS USED

Technology / Tool	Purpose	Learning Outcome
C / C++ / Java fundamentals	Programming and logic development	Improved problem solving and code structure
HTML, CSS and JavaScript	Understanding basic web interfaces	Awareness of frontend development workflow
		Understanding structured data and Understanding
Git / Version Control	Tracking and managing code changes	development collaborative Efficient development workflow
IDE / Code Editor	Writing, running and debugging	

	code	
Issue / Task Tracking	Recording tasks, bugs and progress	Professional task management
Testing Tools / Techniques	Checking correctness and identifying defects	Validation and debugging skills

4.7 SYSTEM ARCHITECTURE

The project workflow can be represented as a simple layered architecture. A requirement or issue is first received at the task level. It is analysed and converted into an implementation plan. The development layer contains the application logic or configuration being worked on. The data layer represents any required database or structured data. Testing and validation are performed before the task is documented and communicated.

A simplified flow is:

Requirement / Issue → Analysis → Development / Configuration → Data Interaction → Testing → Debugging → Documentation → Review / Handover

Figure 4.1: Simplified architecture and workflow of the internship project.

For an actual submission, a diagram created from the intern's real project workflow may be inserted below this paragraph.

4.8 METHODOLOGY

The internship methodology follows an iterative software and IT task workflow. Each task is understood before implementation, and the result is checked before it is considered complete.

Step 1: Requirement Understanding

The first step is to understand the requirement, issue or assigned task. Important information includes the expected output, inputs, constraints, dependencies and acceptance conditions. Clarifying questions are raised when the requirement is incomplete.

Step 2: Task Analysis

The task is divided into smaller technical activities. The intern identifies the relevant application component, data requirement, development tool or troubleshooting area. Breaking a task into smaller units makes it easier to estimate effort and identify possible risks.

Step 3: Planning the Solution

A basic solution approach is prepared before implementation. The plan may include the required code changes, database operations, configuration changes, test cases and documentation. The solution should be consistent with the team's existing practices.

Step 4: Development / Configuration

The required change is implemented in a controlled development environment. Code is written with attention to readability, modularity and maintainability. Where configuration or troubleshooting is involved, changes are made carefully and recorded.

Step 5: Testing and Debugging

The solution is tested against expected behaviour. Errors are reproduced, analysed and corrected. Basic functional testing is performed using normal inputs as well as selected edge cases. The objective is to confirm that the change works and does not introduce an obvious regression.

Step 6: Version Control and Review

Changes are tracked through version-control practices where applicable. Meaningful change descriptions help other team members understand what was modified. Work may be reviewed by a mentor or senior team member before completion.

Step 7: Documentation

The final step is to document the work completed, important observations, test results and any limitations or follow-up actions. Good documentation improves knowledge sharing and makes future maintenance easier.

4.9 EXPECTED OUTCOMES

- Better understanding of professional software and IT workflows.
- Improved programming, debugging and logical problem-solving ability.
- Practical awareness of automobile operations and technology life-cycle activities.
- Basic experience with version control, testing and issue tracking.
- Improved understanding of databases, web technologies and application workflows.
- Improved technical documentation and communication skills.
- Greater awareness of professional ethics, confidentiality and security practices.
- Ability to work more effectively in a team-based automobile industry environment.

4.10 CERTIFICATES OF COMPLETION AND OTHER COMMUNICATION PROOF

The University format requires the internship completion certificate and relevant communication proof to be attached. The following items should be inserted in the final printed report, if applicable:

- Internship Completion Certificate issued by Mahindra Automobile.
- Internship offer / joining communication, if permitted for academic submission.
- Completion or evaluation communication from the internship mentor.
- Any approved project/task acknowledgement or assessment document.

Important: Only documents that are authentic and approved for academic submission should be attached. Personal contact details, credentials, confidential project information and restricted company material should be redacted where necessary.

INTERNSHIP LEARNING SUMMARY

The internship helped me understand how work is carried out in a professional automobile organisation. I learned the importance of following instructions, maintaining records, working with a team and completing tasks carefully.

I also improved my basic technical knowledge and learned how technology and computer-based systems support automobile operations. The experience gave me a better understanding of professional discipline, communication and problem solving.

Overall, the internship was a useful learning experience and gave me a foundation for developing my technical and professional skills further.

Overall, the internship provides a foundation for further learning in full-stack development, software engineering, cloud technologies, DevOps and other areas of computer science. Continued practice through academic projects and independent development can build on the experience gained during the internship.

5. BIBLIOGRAPHY / REFERENCES

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